

Paper:

On-Machine Estimation of Workpiece Deformation for Thin-Structured Parts Machining

Koji Teramoto[†]

Muroran Institute of Technology
27-1 Mizumoto, Muroran, Hokkaido 050-8585, Japan

[†]Corresponding author, E-mail: teramoto@mmm.muroran-it.ac.jp

[Received March 1, 2017; accepted July 11, 2017]

The workholding process in small batch production is one of the least automated processes in machining. In order to ensure appropriate workholding, it is necessary to estimate actual deformation of the workpiece. Recently, near net shape technologies, such as thin-wall casting and additive manufacturing, have become common. Increased requirements for the finish machining of thin-structured parts has increased the need for the appropriateness of workholding to be evaluated. An objective of this study is to investigate an on-machine estimation method that can evaluate the actual deformation of parts with thin-structures. Thin-structured parts are usually held by means of multi-point fixturing or vise fixturing. A hybrid estimation method combining FEM analysis and local strain measurements is adopted to estimate the deformation. The effectiveness of the proposed method is evaluated with example problems. The results indicate the feasibility of the on-machine estimation of the deformation of thin-structured parts.

Keywords: workpiece deformation, workholding, on-machine estimation, thin-structured parts

1. Introduction

In the competitive world of manufacturing, the accurate machining of complicated and thin-structured parts has become an important issue. This trend has been observed not only in the aeronautics industry but also in the consumer products industry.

Advances in CAM systems and CNC controllers have made it possible to operate machine tools with a minimum of human interaction. However, an automatic workholding method for general workpieces has not been developed. Thus, single-purpose jigs are employed to hold workpieces in mass production machining. In small batch machining, it is difficult to prepare such dedicated jigs to hold the workpiece, so workholding in small batch machining is usually executed manually. This sometimes causes workpiece deformation that is beyond the allowable limit for machining error. Therefore, on-machine estimation of actual workpiece deformation is necessary if

reliable workholding is to be achieved.

Because of its low rigidity, the standard workholding of thin-structured parts has become kinematically over-constrained multipoint workholding in order to distribute the loads. Otherwise, mechanical vise fixturing with large contact surfaces is used for ease of operation. These contact situations make the workholding process complex.

A considerable amount of research has been done on workholding over the year [1,2]. Process planning methods considering workholding have also been investigated [3–5]. In order to evaluate the workholding process, numerical simulations of workpiece deformation have been also investigated using the Finite Element Method (FEM) [6–9]. The process simulations of fixturing have been adapted to the planning of deformation-conscious operation planning [10–12]. As an alternative approach to workholding, the development of special devices that are applicable to various low-rigidity parts has also been investigated [13–18]. This approach has drawbacks in terms of its commoditization of facilities, however.

In spite of the energetic research mentioned above, an on-site evaluation of the actual workholding operation has not been performed. Considering the losses involved in machining failures, an evaluation of actual workholding in precision machining is important.

2. Basic Concept

The objective of this research is to develop an on-machine method of estimating workpiece deformation. There is only small spare space on the machine tool. Furthermore, there are continuous vibrations in machining operation. Therefore, the device for on-machine measurement must be small and durable. These requirements indicate that it is important to utilize a minimum number of sensors to estimate workpiece deformation. Recently, many investigations into process estimation, investigations that combine measured data and precise process models, have been reported [19–21]. **Fig. 1** illustrates a method of estimating workpiece deformation by combining a local strain measurement and a three-dimensional elastic analysis. The proposed method consists of a function to estimate adequate boundary conditions of the FEM. The function enables adaptive estimation, which re-

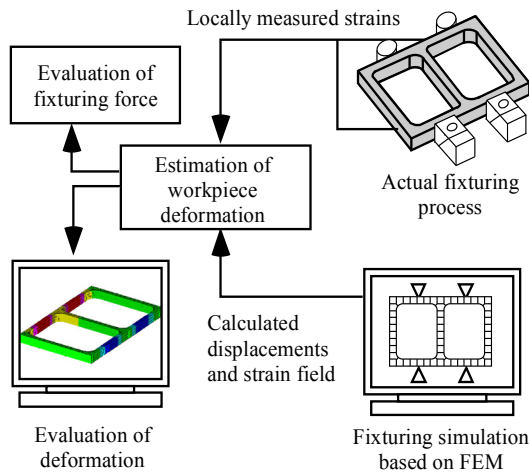


Fig. 1. Framework of on-machine estimation.

flects the variations in actual fixturing situations.

We utilize local strain measurement to estimate the actual workpiece deformation. Elastic strain is only affected by the local behavior of the workpiece. This means that strain-based estimation can separate workpiece deformation information from locating errors. When the workpiece deformation is within the allowable tolerances, only a few points of workpiece surface measurement are required in order to evaluate the position and orientation because the workpiece can be considered as a rigid body. Furthermore, recent research into strain measurement has enabled non-contact measurement of the strain by using laser speckle correlation [22]. This implies the possibility of the utilization of local strain measurement for the on-machine estimation of workpiece deformation.

3. Estimating Workpiece Deformation

3.1. Representation of Workholding State

Based on the standard FEM formulation, the elastic deformation of a solid object is described as follows.

$$\mathbf{F} = \mathbf{K}\mathbf{U} \quad (1)$$

where $\mathbf{F}$ is a nodal force vector, $\mathbf{K}$ is a stiffness matrix, and $\mathbf{U}$ is a nodal displacement vector. In the standard elastic analysis, parts of $\mathbf{F}$ and $\mathbf{U}$ are given as boundary conditions. Assume the force and displacement boundaries are denoted by $\underline{\mathbf{f}}$ and $\underline{\mathbf{u}}$ respectively. $\mathbf{F}$ and $\mathbf{U}$ are decomposed as $\mathbf{F} = \{\underline{\mathbf{f}}, \mathbf{f}\}^T$, $\mathbf{U} = \{\underline{\mathbf{u}}, \mathbf{u}\}^T$. When the boundary conditions $\underline{\mathbf{f}}$ and $\underline{\mathbf{u}}$ are determined, Eq. (1) can be solved through an established procedure.

Concerning the actual fixturing situation, boundary conditions are complicated because the effect of real fixtures, such as friction, slippage, and stress distribution, must be considered [23, 24]. However, even in a precise fixturing simulation based on solid contact mechanics, the distribution of deformation for a known workpiece can be determined by the nodal force vector and the nodal

displacement vector. Therefore, there is a set of nodal displacement distributions and nodal force distributions of boundary constraints which is equivalent to the complex boundary phenomena. This indicates that the standard FEM representation has enough information about the deformation of the workpiece. Thus, the problem can be described by determining equivalent boundary conditions. This means that the workpiece state can be described using three pieces of information: boundary conditions, workpiece shape, and material properties.

Furthermore, because the allowable machining error is small enough for the infinitesimal strain approximation to be introduced, the problem can be modeled using the linear elastic theory. Therefore, it can be assumed that there is an equivalent problem that has simple force and displacement constraints.

Finally, it can be said that the workholding state is represented by workpiece properties and equivalent boundary conditions. Workpiece properties contain both shape and material information. They are used to calculate the stiffness matrix $\mathbf{K}$. The equivalent boundary condition is described as a set of force constraints and displacement constraints.

3.2. Procedure for Estimating Fixturing State

The strain field of a workpiece is calculated based on displacements of the workpiece. Therefore, calculated workpiece strains at measuring points can be described, as follows:

$$\mathbf{e} = \mathbf{G}(\mathbf{U})$$

where $\mathbf{e}$ is a set of calculated strains, $\mathbf{G}()$ is a function determined by the shape function of the related node, and $\mathbf{U}$ represents the calculated displacements of the workpiece. Assume $\mathbf{e}^{\sim}$ is a set of measured strains. The proposed estimation problem can be formalized as follows.

$$\begin{aligned} &\text{find } \underline{\mathbf{u}} \text{ and } \underline{\mathbf{f}} \\ &\text{that minimize } \|\mathbf{e} - \mathbf{e}^{\sim}\| \end{aligned}$$

This is a simple standard optimization problem. However, in order to adapt the method to realistic situations, the measurement of strain must be localized and the number of measurements must be small. Usually, the degree of freedom of $\underline{\mathbf{u}}$ is much larger than the number of measuring points. The number of measuring points is equal to the dimension of $\mathbf{e}^{\sim}$. This means the problem is an ill-posed problem [25]. By introducing an adequate evaluation function, it may be possible to regularize the problem. However, this may be a time-consuming process.

Thus, in order to solve this ill-posed problem, we introduce a heuristic approach called the boundary condition pattern (BCP). The concept of BCP is based on the following two assumptions.

- 1) Characteristic variation cases of boundary conditions can be supposed in advance.
- 2) An actual fixturing situation can be approximated by the weighted linear sum of the supposed conditions.

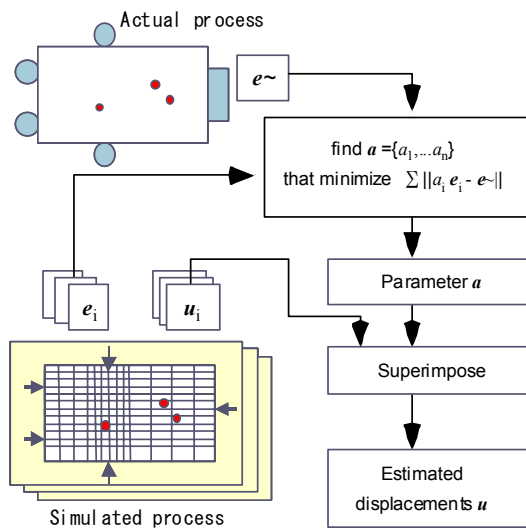


Fig. 2. Process flow of estimation.

Based on the assumptions, the estimation problem is formalized as follows.

$$\begin{aligned} &\text{find } \mathbf{a} = \{a_1, \dots, a_n\} \\ &\text{that minimize } \left\| \sum_{i=1}^n a_i \mathbf{e}_i - \mathbf{e}^{\sim} \right\| \end{aligned}$$

Here $\mathbf{e}_i$ is the calculated strain by the i -th boundary condition pattern ($\mathbf{u}_i, \mathbf{f}_i$), and vector $\mathbf{a}$ is a set of weight parameters for superimposition. In this case, the number of the pattern denoted as “ n ” can be determined by operators. Then, if “ n ” is equal to or smaller than the dimension of $\mathbf{e}$, the problem can be solved using the generalized inverse matrix. A process flow of the estimation is illustrated in Fig. 2. The procedure for estimation can be summarized by the following five steps.

1. Determining the number and position of $\mathbf{e}^{\sim}$.
2. Preparing the boundary condition patterns.
3. Calculating the $\mathbf{e}_i$ and $\mathbf{u}_i$ based on the patterns.
4. Finding the parameter set $\mathbf{a}$.
5. Superimposing displacement fields $\mathbf{u}_i$.

By executing these five steps, an estimated displacement field of a workpiece is acquired. Because this procedure does not contain an iterative sequence, calculation times will be short. Even if the workpiece has a complex shape, the linear elastic analysis calculations will complete within a reasonable time.

4. Boundary Condition Pattern

In order to apply the proposed method to a workholding situation, a systematic preparation of BCPs is necessary. Based on the abovementioned assumptions, appropriate BCPs must satisfy the following conditions:

- A. Every BCP must be able to immobilize the workpiece in order to guarantee stable calculation.
- B. A set of BCPs must cover the effects of all contact points.
- C. Every BCP must generate a different respective strain distribution.

To hold the workpiece there are two popular methods used, a combined clamping set method and a mechanical vise method. These are quite different from the aspect of mechanical analysis. When we employ the clamping set, the boundary can be represented as multipoint constraints. If a mechanical vise is used, the boundary can be represented as two parallel surfaces constraints.

4.1. BCPs for Multipoint Fixturing

As the typical features of using multipoint fixturing to hold low-rigidity parts, the number of contact points between the workpiece and holding devices such as locators and clamps is larger than the number of kinematically sufficient workholding points. Because contact points can be observed easily, fixture location error is not a serious problem. The main workholding error of the multipoint workholding would come from variations in the fixturing force.

Based on the assumption mentioned above, a following preparation procedure of the BCP can be determined.

1. Determining the number of the BCPs based on the number of strain measurements.
2. Extracting the contact situation from the allocation plan of locators and clamps.
3. Modeling the boundary condition for the FEM analysis.
4. Enumerating simplified boundary conditions (SBCs) which satisfy condition A.
5. Determining the set of BCPs by collecting the SBC and/or combined SBCs so that the set satisfies conditions B and C.

Based on the procedure, BCPs for multipoint workholding can be introduced.

4.2. BCPs for Vise Fixturing

Mechanical vises are commonly used in machining. Unlike in multipoint fixturing, there are boundary conditions for two large contact areas in vise fixturing. From the viewpoint of boundary conditions in FEM, boundary nodes within contact areas are the same as contact points in multipoint fixturing. Therefore, it is possible to apply the determination procedures of multipoint fixturing in principle. However, the number of boundary nodes is much larger than the number of contact points for multipoint fixturing. Furthermore, the direction of contact is

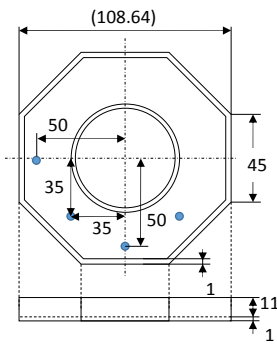


Fig. 3. Dimensions and strain measurement points.

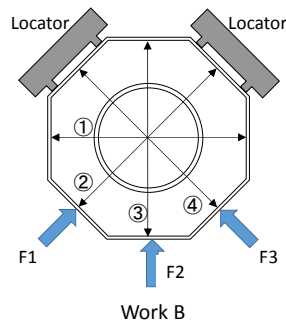


Fig. 4. Workholding situation and dimensions for evaluation.

restricted by the structure of the vise. Therefore, we introduce empirical segmentation of the contact area. Combinations of countered and segmented contact areas that can immobilize the workpiece are extracted as BCPs for vise fixturing. By using countered and segmented contact areas, conditions A to C can be satisfied.

5. Case Studies

5.1. Examples of Multipoint Fixturing

In order to evaluate the proposed method, thin-structured parts were actually fixed as workpieces to serve as examples of estimation. Workholding equipment that can control the fixturing force was employed in the experiments. As an example of a thin-structured part, a thin-walled workpieces was employed. The shape of workpiece, the dimensions, and the strain measurement points are illustrated in **Fig. 3**. Strain gauges (Tokyo Sokki Kenkyujo Co. foil strain gauge FLA-5-23) were employed to measure the strains. **Fig. 4** illustrates the workholding conditions of the workpieces and the dimensions for evaluation. By applying the controlled workholding forces (F1, F2, and F3), variation of dimensions (① to ④) were measured by dial indicators.

Two workholding situations were estimated as follows:

Case A: Estimation in standard workholding situation with multipoint fixturing. Deformations of a workpiece in a symmetrical workholding situation were estimated in order to evaluate basic functionality.

Case B: Estimation in irregular workholding situation with multipoint fixturing. Deformations of a workpiece under the asymmetric workholding load were estimated in order to evaluate adaptability to variations in the fixturing force.

As an example of standard workholding situations, F1, F2 and F3 were controlled as follows (these values were only used for FEM analysis):

$$F1 = F2 = F3 = 50 \text{ N}$$

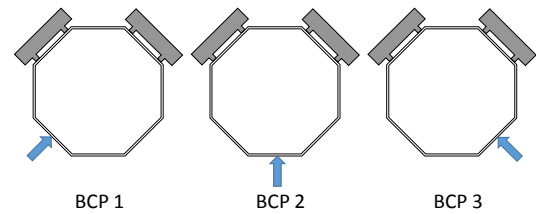


Fig. 5. Determined BCPs.

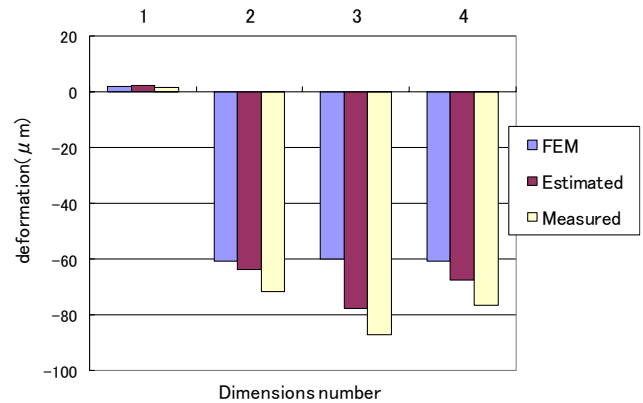


Fig. 6. Comparison of deformations (Case A).

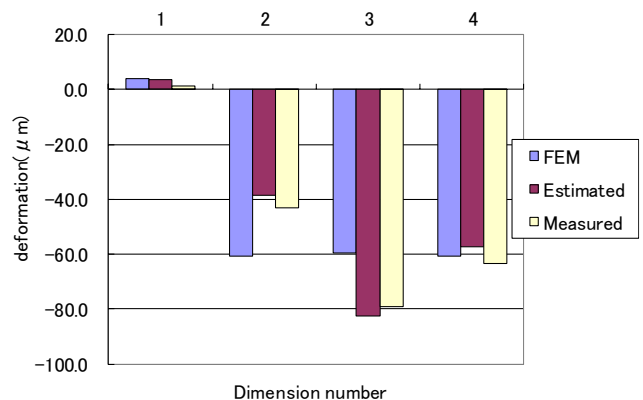


Fig. 7. Comparison of deformations (Case B).

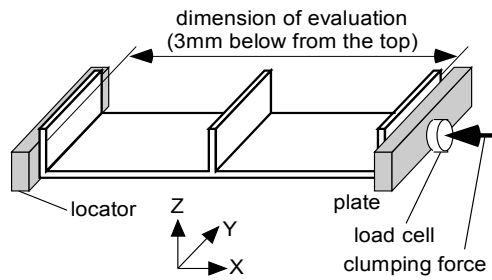
Based on the proposed procedure, BCPs were determined. A schematic diagram of BCPs is shown in **Fig. 5**. Every BCP and the set can satisfy conditions A to C.

In **Fig. 6**, nominal deformations calculated by FEM, estimation results, and measured deformations are compared. The results indicate that the proposed method can produce a more accurate estimate than nominal FEM analysis.

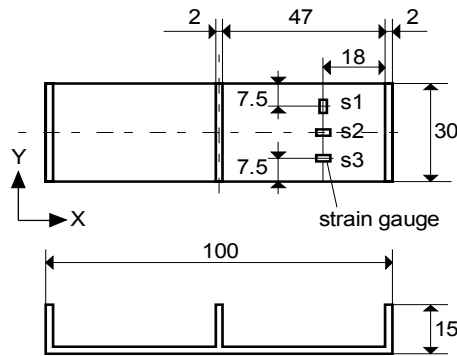
For the estimation of a more realistic workholding situation, asymmetric fixturing forces were loaded. The estimation in this situation was expected to demonstrate the adaptation of fixturing force error. In this case, F1, F2, and F3 were controlled as follows:

$$F1 = 30 \text{ N}, F2 = 50 \text{ N}, F3 = 40 \text{ N}$$

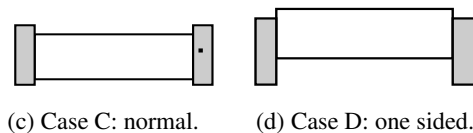
The BCPs in the Case A example were applied. **Fig. 7** compares nominal deformations calculated by FEM (F1 = F2 = F3 = 50 N), estimation results, and measured



(a) Experimental setup for vise fixturing.



(b) Workpiece shape for vise fixturing.

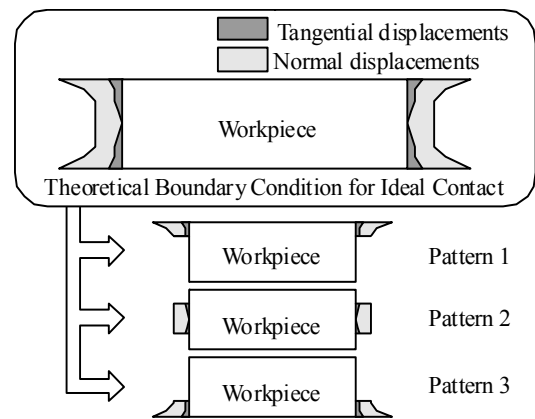
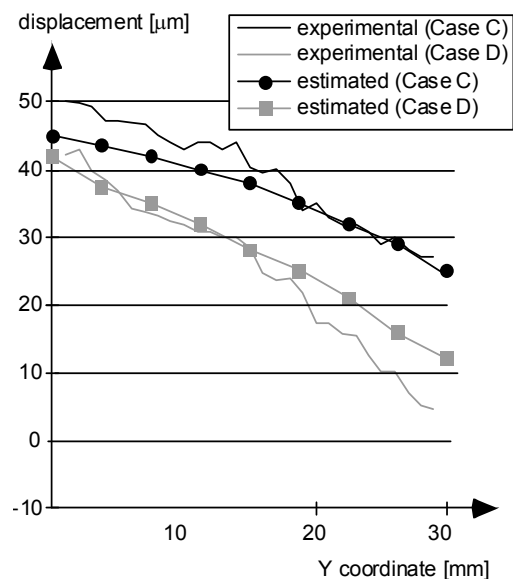
**Fig. 8.** Experimental setup and fixturing conditions.

deformations. The results show that the proposed estimation method can estimate the deformation accurately, even in an unexpected fixturing situation. In an actual workholding situation, it is difficult to have accurate control of the fixturing force. This estimation ability indicates that undesirable fixturing variations can be detected by using the proposed method.

5.2. Examples of Vise Fixturing

Figure 8 illustrates an example problem for vise fixturing. The experimental setup for simulating the vise fixturing is illustrated in **Fig. 8(a)**. The accuracy of estimation is evaluated by comparing the dimensions indicated in **Fig. 8(a)**. The workpiece shape and the strain measurement points are illustrated in **Fig. 8(b)**. The workpiece material is aluminum alloy. There are three strain gauges on the workpiece. Strain gauge s1 is attached in the Y direction; strain gauges s2 and s3 are attached in the X direction. To evaluate the adaptation to an actual situation, two workholding situations were estimated.

Case C: Estimation for a standard workholding situation with vise fixturing. Deformations of a workpiece under normal workholding conditions were estimated in order to evaluate basic functionality.

**Fig. 9.** Determined BCPs for vise fixturing.**Fig. 10.** Comparison of estimated deformation and measured deformation.

Case D: Estimation for an irregular workholding situation with vise fixturing. Deformations of a workpiece under the asymmetric workholding load conditions were estimated in order to evaluate adaptability to variations in fixturing location.

The differences between the two cases are illustrated in **Figs. 8(c)** and **(d)**. Because the parallelism of the workpiece is not reliable, the fixturing force was loaded unequally. A load cell (Kyowa LMB-A-2KN) was used only to equalize the trials. The fixturing force value was not used in the estimation process.

Three BCPs were determined considering a conventional elastic analysis [26] and a kinematic evaluation. Determined BCPs are illustrated in **Fig. 9**.

Figure 10 shows the dimension displacements in **Fig. 8(a)**. Displacements were measured by a three dimensional coordinate measurement machine. Measured displacements and estimated displacements were

expressed. The results show good quantitative agreements in both cases. These results show that the proposed method can accurately estimate the state of the workpieces changes occur in the fixturing situation.

6. Conclusions

In order to evaluate the actual workholding states in small batch machining of thin-structured parts, a method of estimating elastic deformation has been proposed. The proposed method is intended to realize on-machine estimation of workpiece deformation. In particular, the method is expected to support the workholding of low rigidity parts. A hybrid method combining local strain measurement and FEM analysis is introduced to estimate actual workpiece deformation. The estimation method can be understood as the adjustment of numerical elastic analysis using the results of local strain measurements. The effectiveness of the proposed estimation method has been demonstrated by evaluating examples of fixed thin-structured workpieces.

Applying a non-contact strain measurement method will be an important subject of future research. Furthermore, the systematic selection of measurement points will also be very important.

Acknowledgements

The author thanks Mr. Wataru Konno and Mr. Shintaro Ono for their technical assistance. As this work was partly supported by Mazak Foundation and KAKENHI Grant Number 16760092, the author would also like to express his appreciation for their financial aids.

References:

- [1] H. Asada and A. By, "Kinematic Analysis of Workpart Fixturing for Flexible assembly with Automatically Reconfigurable Fixtures," IEEE J. of Robotics and Automation, Vol.RA-1, No.2, pp. 86-94, 1985.
- [2] S. Hirai et al., "A Model-Based Generation of Fixture Layout Candidates for Workpiece Holding," Proc. of 5th Japan-U.S.A. Symposium on Flexible Automation, pp. 595-602, 1994.
- [3] J. R. Boerma and H. J. J. Kals, "FIXES, A System for Automatic Selection of Set-ups and Design of Fixtures," Annals of the CIRP, Vol.37, No.1, pp. 443-446, 1988.
- [4] A. Y. C. Nee and S. Kumar, "A Framework for an Object/ Rule-Based Automated Fixture Design System," Annals of CIRP, Vol.40, No.1, pp. 147-151, 1990.
- [5] H. Sakurai, "Automatic Setup Planning and Fixture Design for Machining," J. of Manufacturing System, Vol.11, No.1, pp. 30-37, 1992.
- [6] J. D. Lee and L. S. Haynes, "Finite Element Analysis of Flexible Fixturing System," J. of Engineering for Industry, Vol.109, pp. 395-406, 1987.
- [7] S. Ratchev et al., "Milling error prediction and compensation in machining of low-rigidity parts," Int. J. of Machine Tools and Manufacture, Vol.44, Issue 15, pp. 1629-1641, December 2004.
- [8] S. P. Siebenaler and S. N. Melkote, "Prediction of workpiece deformation in a fixture system using the finite element method," Int. J. of Machine Tools and Manufacture, Vol.46, No.1, pp. 51-58, 2006.
- [9] J. Wang, S. Ibaraki, A. Matsubara, K. Shida, and T. Yamada, "FEM-Based Simulation for Workpiece Deformation in Thin-Wall Milling," Int. J. of Automation Technology, Vol.9, No.2, pp. 122-128, 2015.
- [10] W. Chai et al., "Deformable Sheet Metal Fixturing : Principles, Algorithms, and Simulations," J. of Manufacturing Science and Engineering, Vol.118, pp. 318-324, Aug. 1996.
- [11] D. Freiburg et al., "Simulation based Process Optimization for the Milling of Light Weight Components," Procedia CIRP, Vol.18, pp. 132-137, 2014.
- [12] Z. Cai et al., "Systematic Solving of Machining Deformation and Process Optimization for Complex Thin-walled Parts," Procedia CIRP, Vol.56, pp. 167-172, 2016.
- [13] S. E. Sarma and P. K. Wright, "Reference Free Part Encapsulation: A new universal fixturing concept," J. of Manufacturing Systems, Vol.16, No.1, pp. 35-47, 1997.
- [14] H. Obara et al., "A Method to Machine Three-Dimensional Thin Parts," J. of the Japan Society for Precision Engineering, Vol.69, No.3, pp. 375-379, 2003 (in Japanese).
- [15] J. F. Hurtado and S. N. Melkote, "A model for synthesis of the fixturing configuration in pin-array type flexible machining fixtures," Int. J. of Machine Tools and Manufacture, Vol.42, No.7, May, pp. 837-849, 2002.
- [16] E. C. D. Meter, "Light Activated Adhesive Gripper (LAAG) Workholding Technology and Process," J. of Manufacturing Processes, Vol.6, No.2, pp. 201-214, 2004.
- [17] T. Aoyama and Y. Kakinuma, "Development of Fixture Devices for Thin and Compliant Workpieces," Annals of the CIRP, Vol.54, No.1, pp. 325-328, 2005.
- [18] K. Nakamoto et al., "Dexterous Machining of Soft Objects by Means of Flexible Clamper," Int. J. of Automation Technology, Vol.9, No.1, pp. 83-88, 2015.
- [19] K. Teramoto et al., "Thermal State Visualization of Machining Workpiece by Means of a Sensor-Configured Heat Conduction Simulation," JSME Int. J. Series C, Vol.49, No.2, pp. 287-292, 2006.
- [20] T. Ohashi et al., "In-Process Measurement of Elastic Deformation of a Large Deep-Drawing-Die with Fusion of Experiment and Numerical Analysis," Int. J. of Automation Technology, Vol.3, No.4, pp. 457-464, 2009.
- [21] M. Baeker, "A new method to determine material parameters from machining simulations using inverse identification," Procedia CIRP, Vol.31, pp. 399-404, 2015.
- [22] I. Yamaguchi, "Displacement and Strain Measurement by Electronic Speckle Correlation," Oyo Butsuri, Vol.69, No.6 pp. 584-587, 1992 (in Japanese).
- [23] K. L. Johnson, "Contact Mechanics," Cambridge University Press, 1987.
- [24] S. Satyanarayana and S. N. Melkote, "Finite element modeling of fixture - workpiece contacts: single contact modeling and experimental verification," Int. J. of Machine Tools and Manufacture, Vol.44, No.9, pp. 903-913, 2004.
- [25] S. Kubo (Ed.), "Inverse Problems," Atlanta Technology Pub., 1993.
- [26] T. Tsumura (Ed.), "Contact theory, Data book for strength design," Shokabo, 1971 (in Japanese).



Name:

Koji Teramoto

Affiliation:

Professor, Division of Production Systems Engineering, Muroran Institute of Technology

Address:

27-1 Mizumoto, Muroran, Hokkaido 050-8585, Japan

Brief Biographical History:

1993- Research Associate, Osaka University

2005- Lecturer, Osaka University

2006- Associate Professor, Muroran Institute of Technology

2014- Professor, Muroran Institute of Technology

Main Works:

- "A framework of compositional machining simulation for versatile machining simulation," JSME J. of Advanced Mechanical Design, Systems, and Manufacturing, Vol.2, No.4, pp. 668-674, 2008.

Membership in Academic Societies:

- Japan Society of Mechanical Engineers (JSME)
- Japan Society for Precision Engineering (JSPE)